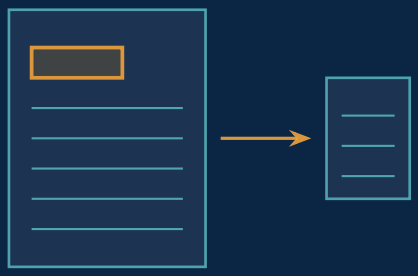
ICT UNIVERSITY · FACULTY OF ICT · DEPARTMENT OF COMPUTER SCIENCE



DocuMind AI

Intelligent Document Processing and Automated ERP Ingestion System

Phase 1 – Requirements Analysis and Planning

REQUIREMENTS SPECIFICATION · SCOPE · RISKS · PROJECT PLAN

Student	Tuheu Tchoubi Pempeme Moussa Fahdil
Programme	B.Sc. Computer Science, final-year project
Supervisor	[Supervisor name – to be completed]
Institution	The ICT University, Cameroon
Version	2.0 – October 2026

Document Control

Item	Description
Document	DocuMind AI – Phase 1: Requirements Analysis and Planning.
Version	2.0 (October 2026).
Deliverable	Requirements specification: problem analysis, objectives, scope, stakeholders, functional and non-functional requirements, use cases, data requirements, success metrics, risks, project plan and traceability.
Related document	*Project Design Document* (architecture, UML, data and API design).
Status	**Draft for supervisor review.** Items marked *[Decision needed]* must be confirmed before Phase 1 is closed (Section 14).
Honesty note	Targets in this document are planning values. No result is claimed.

Contents

Document Control i

1 Purpose and Summary 1

1.1 Purpose of this document . . . 1

1.2 Project in one paragraph . . . 1

1.3 Phase 1 at a glance . . . 1

2 Problem Analysis 2

2.1 Current situation (as-is) . . . 2

2.2 Problems identified . . . 2

2.3 Proposed solution (to-be) . . . 2

2.4 System context . . . 3

3 Objectives and Success Criteria 4

3.1 Aim . . . 4

3.2 Objectives with measurable success criteria . . . 4

3.3 Provisional quantitative targets . . . 4

4 Scope 6

4.1 In scope (first prototype) . . . 6

4.2 Out of scope . . . 6

4.3 Supported document types: decision . . . 6

5 Stakeholders and Users 7

5.1 Stakeholder analysis . . . 7

5.2 User roles . . . 8

5.3 User stories . . . 8

6 Functional Requirements 9

6.1 Requirement list with acceptance criteria . . . 9

6.2 Invoice field specification . . . 10

7 Non-Functional Requirements 12

7.1 Business rules . . . 12

8 Use Cases 13

8.1 Use-case diagram . . . 13

8.2 Detailed use-case specifications . . . 13

9 Data Requirements **15**
9.1 Data the system stores . . . 15
9.2 Dataset requirements for OCR (Phase 8 preparation) . . . 15
9.3 Ethics and data handling . . . 15

10 Constraints, Assumptions and Dependencies **16**
10.1 Constraints . . . 16
10.2 Assumptions . . . 16
10.3 Dependencies . . . 16

11 Risk Analysis **17**

12 Project Plan **19**
12.1 Phases and stages . . . 19
12.2 Proposed timeline . . . 19
12.3 Milestones . . . 19
12.4 Definition of done . . . 20

13 Traceability Matrix **21**

14 Open Decisions and Sign-off **22**
14.1 Decisions needed to close Phase 1 . . . 22
14.2 Phase 1 completion checklist . . . 22
14.3 Approval . . . 22

List of Figures

2.1 Manual process (as-is) compared with the proposed workflow (to-be). 2
2.2 Context diagram: the system, its users and external systems. 3

5.1 Influence–interest grid (initial judgement; adjust with the supervisor). 7

6.1 Anatomy of an invoice (synthetic example) with the fields to extract. 10

8.1 Use cases and actors (all require sign-in). 13

11.1 Initial risk matrix. Identifiers refer to the table below. 17

12.1 Proposed Gantt chart (colours: teal = foundation, amber = intelligence, violet = application, red = delivery). 19

1

Purpose and Summary

1.1 Purpose of this document

Phase 1 answers four questions before any serious development starts: *what problem are we solving, for whom, within which limits, and how will we know we succeeded?* The answers are written as testable requirements so that design (Phase 2), implementation and evaluation can all be traced back to them.

1.2 Project in one paragraph

DocuMind AI receives supplier invoices (PDF, PNG or JPEG) in French or English, improves image quality, recognises the text with a custom-trained OCR model compared against an established baseline, extracts the key invoice fields, validates them with business rules, asks a human reviewer to confirm uncertain values, and finally sends approved records to an ERP system (a mock or sandbox first, Odoo as target) with a complete audit trail.

1.3 Phase 1 at a glance

Topic	Position taken in this document
Document type	Invoices only for the prototype; other documents are later extensions.
Languages	French and English.
Input formats	PDF, PNG, JPEG.
Human review	Mandatory before any ERP submission.
ERP target	Mock ERP first; Odoo when access and version are confirmed.
OCR	Custom CRNN + CTC model, evaluated honestly against Tesseract.
Architecture	Modular monolith first, then Docker Compose, then services and Kubernetes.
Requirements	22 functional and 10 non-functional requirements with acceptance criteria.

2

Problem Analysis

2.1 Current situation (as-is)

Invoices arrive by e-mail, on paper or as scans. An employee reads each one and types the values into the ERP by hand, usually without a systematic check and without a record of who changed what.

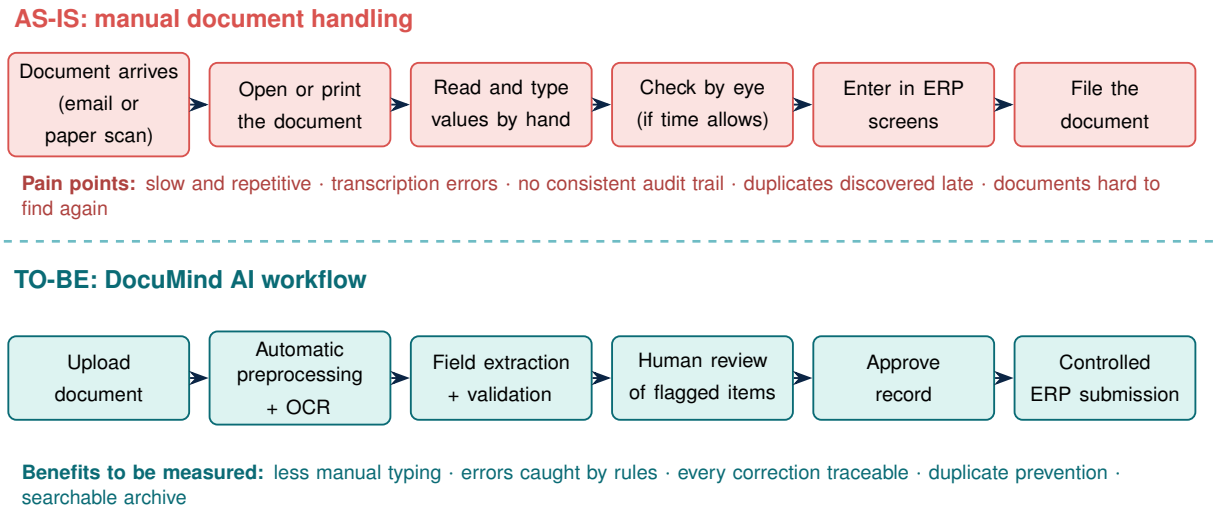


Figure 2.1: *Manual process (as-is) compared with the proposed workflow (to-be).*

2.2 Problems identified

ID	Problem	Consequence
P1	Manual re-typing	Slow, repetitive work that does not scale with volume.
P2	Transcription errors	Wrong amounts, dates or supplier data enter the ERP.
P3	No systematic validation	Totals that do not match subtotal plus tax go unnoticed.
P4	Duplicate invoices	The same invoice may be paid twice.
P5	No audit trail	Corrections cannot be traced to a person or a reason.
P6	Documents hard to retrieve	Originals are scattered across mailboxes and folders.

2.3 Proposed solution (to-be)

Automate the repetitive steps (reading, extracting, cross-checking) and keep the human where judgement matters (resolving uncertainty, approving). Every document and every correction is stored with identifiers and timestamps.

2.4 System context

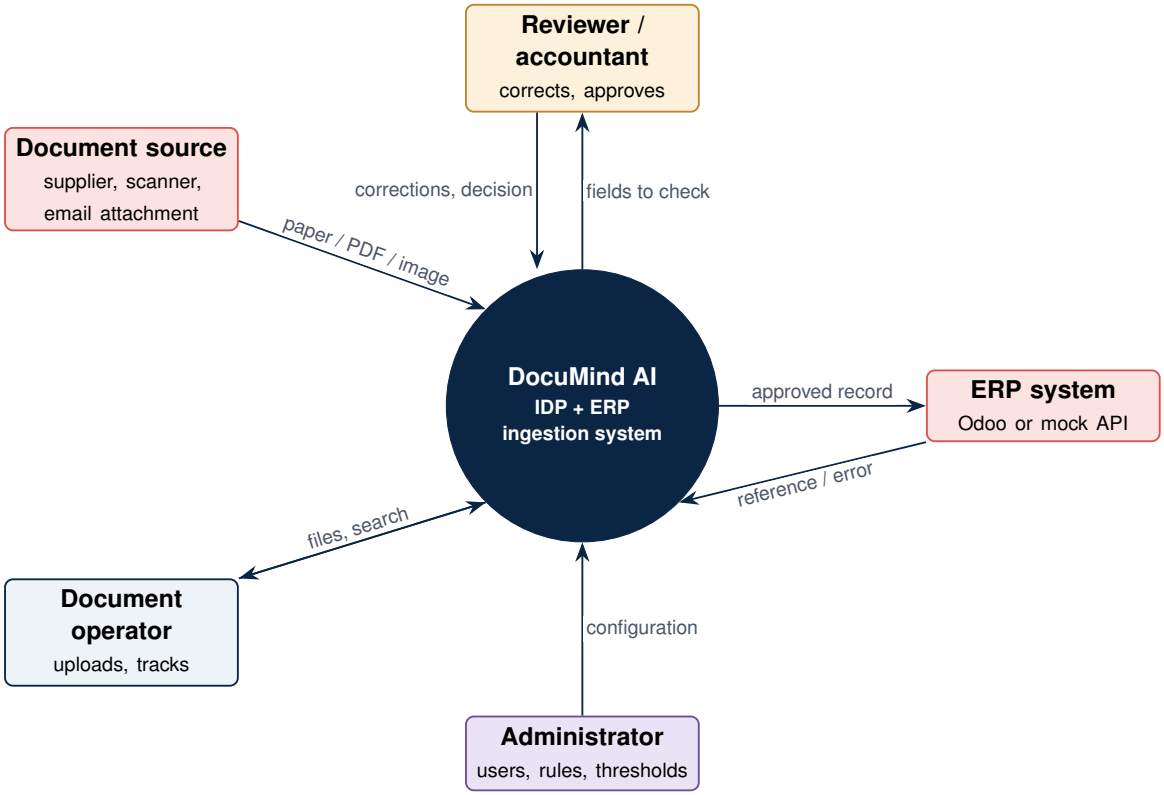


Figure 2.2: *Context diagram: the system, its users and external systems.*

3

Objectives and Success Criteria

3.1 Aim

To design and implement a secure, modular system that extracts structured information from French and English invoices and prepares validated records for ERP ingestion.

3.2 Objectives with measurable success criteria

ID	Objective	Success criterion (to be verified in Phase 14)
O1	Secure upload and lifecycle tracking	A valid invoice receives an ID and persistent status; invalid files are refused with a clear reason.
O2	Improve scan readability	Preprocessing reduces CER on the test set versus no preprocessing (reported, not assumed).
O3	Custom OCR for French and English	CER and WER reported on a held-out set alongside the Tesseract baseline.
O4	Extract key invoice fields	Per-field precision, recall and F1 reported for the field list in Section 6.2.
O5	Confidence and review routing	Flagged records and their error rates reported; thresholds chosen on validation data only.
O6	Validate extracted data	Seeded error cases (wrong total, bad date, duplicate) are detected in tests.
O7	Review dashboard	A reviewer completes correct-and-approve tasks in a usability check.
O8	Persist data and audit trail	Every correction, approval and ERP attempt appears in the audit log.
O9	ERP integration	Mock/sandbox demonstrates success, failure and retry with zero duplicate records.
O10	Containerised delivery	A new machine runs the system from the documented setup in a clean environment.

3.3 Provisional quantitative targets

Planning values, not results

The numbers below only give the experiments a goal. They will be revised after the baseline measurement (Phase 7) and must never be quoted in the report as achieved results.

Metric	Provisional target	Revision point
CER / WER (custom model)	Compete with the Tesseract baseline on the same test set	Absolute target set after Phase 7
Field F1: invoice number, date, total	≥ 0.85 before human review	After Phase 9 first run
Arithmetic-error detection	All seeded error cases detected	Phase 10
Duplicate ERP submissions	0 in retry tests	Phase 12
Median time per single-page invoice	Reported (no fixed target)	After Phase 7
Upload rejection of invalid files	100% of defined invalid test files	Phase 5

4

Scope

4.1 In scope (first prototype)

- Supplier invoices as PDF, PNG or JPEG, French and English, printed or cleanly scanned.
- Preprocessing, OCR (custom and baseline), extraction, validation, confidence scoring.
- Human review with correction, approval, rejection and audit trail.
- ERP adapter with a mock or sandbox ERP; Odoo mapping documented.
- React web dashboard, FastAPI backend, MySQL, Docker.

4.2 Out of scope

- Handwritten invoices and very low-quality photographs.
- Other document types (purchase orders, receipts, forms) – listed as future work.
- Fully automatic posting without human approval.
- Production ERP connection and any claim of legal or accounting compliance.
- Mobile application and offline mode.

4.3 Supported document types: decision

Document type	Prototype?	Reason
Invoice	**Yes**	Clear field set, directly useful for ERP ingestion, arithmetic gives strong validation rules.
Purchase order	Later	Similar fields; add after the invoice workflow works.
Receipt	Later	Variable layouts and thermal-print quality make OCR harder.
Administrative form	Later	Needs form-specific templates.

5

Stakeholders and Users

5.1 Stakeholder analysis

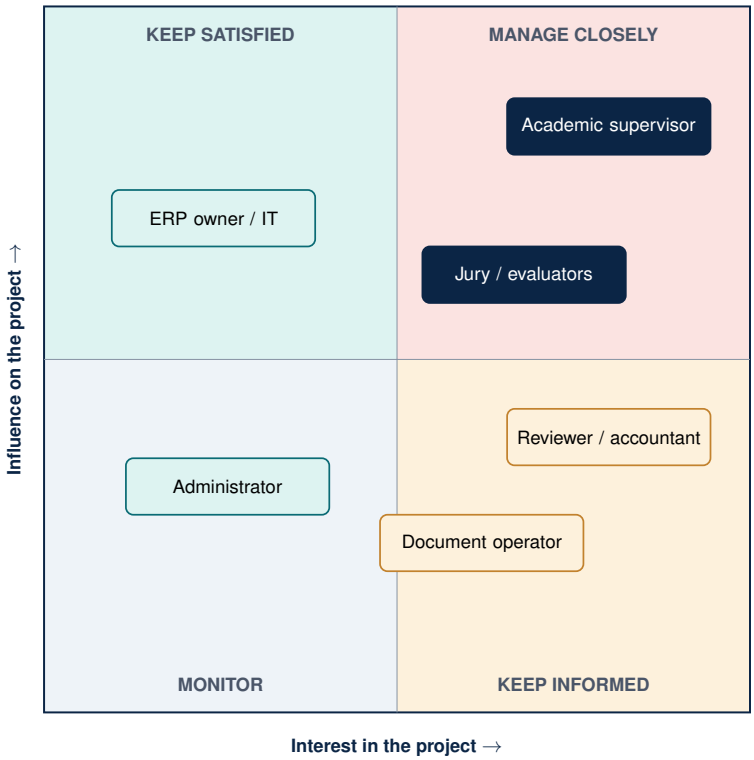


Figure 5.1: *Influence–interest grid (initial judgement; adjust with the supervisor).*

Stakeholder	Needs	Involvement
Academic supervisor	Rigour, honest evaluation, progress evidence.	Reviews documents and milestones.
Jury / evaluators	Clear architecture, working demonstration, measured results.	Final defence.
Reviewer / accountant	Fast, trustworthy correction of uncertain fields.	Main interactive user of the review screen.
Document operator	Simple upload and status tracking.	Daily user.
Administrator	Control of users, rules and thresholds.	Configuration and oversight.
ERP owner / IT	Clean, safe, non-duplicating integration.	Provides interface details (open decision).

5.2 User roles

Four roles are enforced by the backend: **Administrator**, **Operator**, **Reviewer** and **Integration service account**. A supervisor or auditor receives read-only access to logs and metrics.

5.3 User stories

ID	Story	Req.
US-01	As an operator, I upload an invoice and immediately get an ID and status, so that I know it was received.	FR-03, 05, 06
US-02	As an operator, I see why a bad file was refused, so that I can send a correct one.	FR-04
US-03	As a reviewer, I see the source page next to the extracted fields with doubtful values highlighted, so that I check only what matters.	FR-11, 22
US-04	As a reviewer, I correct a value and give a reason, so that the change is traceable.	FR-12, 13
US-05	As a reviewer, I approve or reject a record, so that only checked data reaches the ERP.	FR-12, 14
US-06	As an administrator, I adjust review thresholds and validation rules without changing code.	FR-17
US-07	As an accountant, I am warned about a probable duplicate invoice before it is posted.	FR-10, 11
US-08	As an auditor, I see who changed what and when.	FR-13
US-09	As an operator, I resubmit a failed document for processing.	FR-21
US-10	As the ERP owner, I need a failed submission to be retryable without creating a second record.	FR-15

6

Functional Requirements

6.1 Requirement list with acceptance criteria

Priorities follow MoSCoW: **Must** (needed for the prototype), **Should** (important, can slip), **Could** (nice to have). Each requirement has an acceptance criterion that becomes a test in Phase 14.

ID	Requirement	Prio.	Acceptance criterion
FR-01	A permitted user can sign in and sign out.	Must	Valid credentials give access; invalid ones are refused; logout invalidates the session.
FR-02	Role-based access to admin, operational and review functions.	Must	Each role can call only its endpoints; others return 403.
FR-03	Upload PDF, PNG or JPEG within configured limits.	Must	Valid file returns a document ID.
FR-04	Check file type, size and basic integrity.	Must	Oversized, wrong-type and corrupt files are refused (413/415/400).
FR-05	Store the original privately with a unique ID and SHA-256.	Must	File is not reachable without authorisation; hash stored.
FR-06	Create a processing job with status and timestamps.	Must	Status and times visible after upload.
FR-07	Normalise pages and enhance images.	Must	Pages are rendered and preprocessed; intermediates retrievable in debug mode.
FR-08	OCR returns text and confidence where supported.	Must	Output contains text per line with a confidence value.
FR-09	Map recognised content to configured invoice fields.	Must	Field list in Section 6.2 produced for a test invoice.
FR-10	Validate required fields, formats, arithmetic and duplicates.	Must	Seeded invalid invoices raise the expected findings.
FR-11	Flag low-confidence or invalid records for review.	Must	Flagged records enter the review queue.
FR-12	Reviewer can correct, give a reason, approve or reject.	Must	Decision and reason stored; state changes accordingly.
FR-13	Audit trail of corrections, decisions and ERP attempts.	Must	Each action yields an audit event with actor and time.
FR-14	Only approved records go to the ERP adapter.	Must	Submitting an unapproved record returns 409.
FR-15	Record ERP status, reference and errors; no duplicates.	Must	Repeating a successful submission creates no second ERP record.
FR-16	Search and filter documents.	Should	Filters by status, date, type and ID return correct sets.
FR-17	Admin configures rules and thresholds.	Should	Changed threshold changes review routing without redeploying.
FR-18	Health endpoints and basic metrics.	Should	`/api/v1/health` answers; database readiness reported.

ID	Requirement	Prio.	Acceptance criterion
FR-19	Dashboard shows counts, failure and review queues.	Should	Counts match database state.
FR-20	Export approved data (JSON/CSV).	Could	Export file matches approved records.
FR-21	Resubmit failed or rejected document for reprocessing.	Should	A new run is created; history of earlier runs is kept.
FR-22	Review screen shows source beside fields with highlighted evidence.	Should	Selecting a field highlights its region on the page.

6.2 Invoice field specification

The prototype extracts the fields below. The numbers match the callouts in Figure 6.1. ERP mappings are candidates for a vendor bill in Odoo and **must be verified against the chosen Odoo version** before implementation.

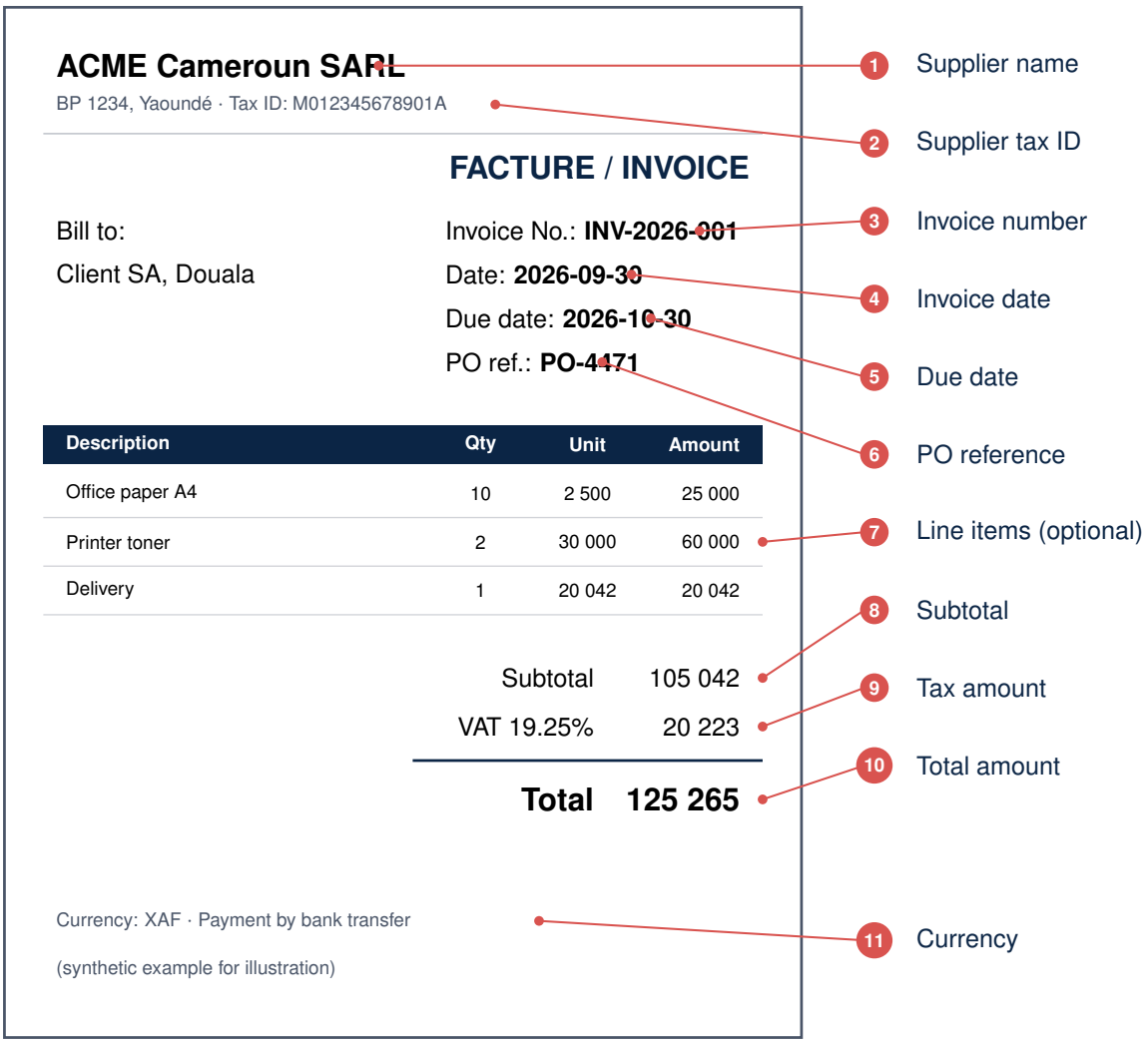


Figure 6.1: *Anatomy of an invoice (synthetic example) with the fields to extract.*

No.	Field	Type	Required	Validation	Candidate ERP field
1	Supplier name	text	Yes	Non-empty; matched to known suppliers if available.	partner
2	Supplier tax ID	text	No	Pattern check where format is known.	partner VAT

No.	Field	Type	Required	Validation	Candidate ERP field
3	Invoice number	text	Yes	Non-empty; used in duplicate check.	vendor reference
4	Invoice date	date	Yes	Parses; plausible range.	invoice date
5	Due date	date	No	Not before invoice date.	due date
6	PO reference	text	No	Format check if configured.	origin / purchase order
7	Line items	table	No	Quantity $\times$ unit price $\approx$ amount.	invoice lines
8	Subtotal	decimal	Yes	Parses with locale rules.	untaxed amount
9	Tax amount	decimal	Yes	Parses; consistent with rate if known.	tax amount
10	Total amount	decimal	Yes	Subtotal + tax $\approx$ total within tolerance.	total amount
11	Currency	code	Yes	Known currency code or symbol; consistent.	currency

7

Non-Functional Requirements

ID	Category	Requirement	Verification
NFR-01	Security	Authenticate users, authorise every protected action, keep secrets out of code, restrict stored files.	Access-control and file-access tests.
NFR-02	Privacy	Collect only necessary data; synthetic or authorised test documents; define retention.	Data-handling checklist.
NFR-03	Reliability	A failed step produces a visible, recoverable state; no silent job loss.	Failure-injection tests.
NFR-04	Traceability	Stable identifiers and timestamps for documents and runs.	Inspection of records.
NFR-05	Maintainability	Separate API, domain logic, OCR, validation, storage and ERP adapter.	Code structure review.
NFR-06	Portability	Setup reproducible from documentation and Docker images.	Clean-machine setup test.
NFR-07	Performance	Measure per-stage and end-to-end time; set targets after a baseline.	Timed runs on the test set.
NFR-08	Usability	Reviewers find doubtful fields and understand errors without logs.	Task-based usability check.
NFR-09	Scalability	OCR separable from interactive requests; scalable later.	Architecture review, worker test.
NFR-10	Observability	Structured logs and health without sensitive content.	Log inspection.

7.1 Business rules

BR-1 No ERP submission before validation and approval by an authorised reviewer.
BR-2 Low-confidence fields are always visible to the reviewer; confidence is not proof.
BR-3 Likely duplicates are flagged, never deleted automatically.
BR-4 ERP requests carry an idempotency key.
BR-5 Corrections keep the original value, new value, actor, time and reason.
BR-6 Failed integration attempts are retryable without duplicate records.

8

Use Cases

8.1 Use-case diagram

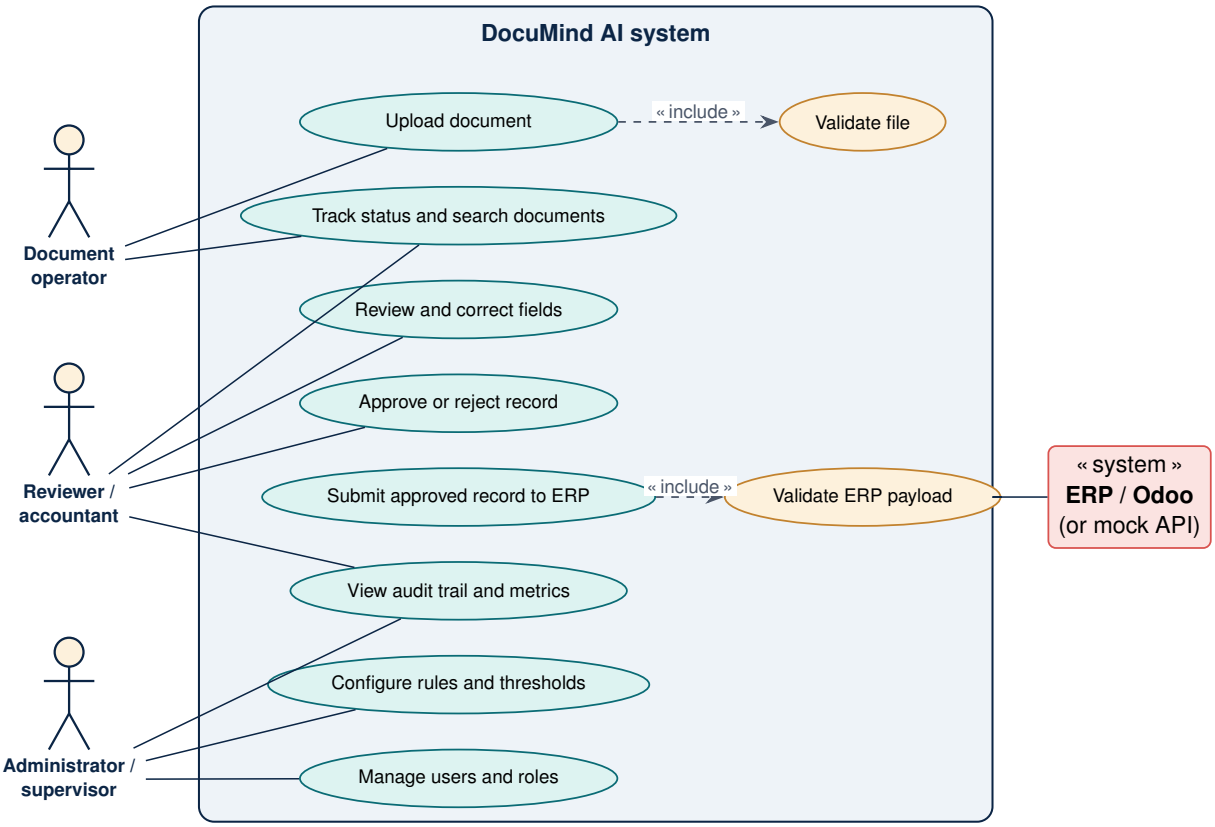


Figure 8.1: *Use cases and actors (all require sign-in).*

8.2 Detailed use-case specifications

UC-01 Upload document

Actor	Document operator
Preconditions	Operator is signed in and has the Operator role.
Main flow	1. Operator selects a file. 2. System checks type, size and integrity. 3. System stores the original and its SHA-256. 4. System creates a document and a queued job. 5. System returns the document ID and status.
Alternative flows	2a. File invalid: system refuses with a reason (400/413/415). 3a. Same hash already stored: system accepts but flags a probable duplicate.

Postconditions	Document exists in state QUEUED; an audit event *upload* is recorded.
Requirements	FR-03, FR-04, FR-05, FR-06, FR-13

UC-02 Review, correct and approve

Actor	Reviewer
Preconditions	Document is in NEEDS_REVIEW.
Main flow	1. Reviewer opens the record. 2. System shows the source page beside extracted fields, doubtful ones highlighted. 3. Reviewer corrects fields and gives reasons. 4. Reviewer approves. 5. System stores corrections and decision and sets APPROVED.
Alternative flows	4a. Reviewer rejects with a reason: state REJECTED. 3a. A correction breaks a validation rule: system shows the finding and blocks approval until resolved or explicitly overridden with a reason (override policy to be decided).
Postconditions	Audit events for each correction and the decision.
Requirements	FR-11, FR-12, FR-13, FR-22

UC-03 Submit approved record to ERP

Actor	Integration service (triggered by an authorised user or automatically after approval)
Preconditions	Document is APPROVED; ERP adapter configured.
Main flow	1. System builds the ERP payload and an idempotency key. 2. System validates the payload. 3. System calls the ERP. 4. On success it stores the external reference and sets ERP_SENT.
Alternative flows	2a. Payload invalid: submission blocked, finding shown. 3a. ERP error or timeout: attempt recorded, state ERP_FAILED; retry reuses the same key.
Postconditions	An ERPAttempt record exists for every try; no duplicate ERP record.
Requirements	FR-14, FR-15, FR-13

UC-04 Manage users and thresholds

Actor	Administrator
Main flow	Create or deactivate users, assign roles, adjust validation rules and review thresholds; every change is audited.
Requirements	FR-02, FR-17, FR-13

9

Data Requirements

9.1 Data the system stores

Users and roles; documents (metadata, hash, storage key, status); processing runs; extracted fields with confidence and evidence; reviews; ERP attempts; audit events. Original files are stored outside the database in protected storage.

9.2 Dataset requirements for OCR (Phase 8 preparation)

- **Sources:** synthetic invoices generated from templates (with varied fonts, layouts and noise), invoices created by the student, and authorised real samples with personal data removed. Do not use real customer data without permission.
- **Languages:** French and English, including accented characters, currency symbols and typical French number formats.
- **Labels:** verified transcriptions of text lines and field-level ground truth for the test set.
- **Split:** training, validation and test sets separated by source document or template, so the test set is truly unseen.
- **Documentation:** dataset version, size, language mix, labelling method and known limitations.
- **Size:** not fixed in advance; decide after the baseline shows the error level. [Decision needed with supervisor]

9.3 Ethics and data handling

Documents may contain personal and financial information. Use synthetic or anonymised data for development, restrict access, define retention, and do not send documents to external OCR services without understanding the privacy implications.

10

Constraints, Assumptions and Dependencies

10.1 Constraints

- One student developer and a fixed academic calendar.
- Limited training hardware; a GPU may need a cloud notebook.
- The ERP interface may be unavailable; a mock ERP is the fallback.
- The jury expects measured results and a working demonstration.

10.2 Assumptions

- Python 3.13 and MySQL 8 can be installed on the development machine.
- Enough invoices (real, synthetic or both) can be assembled for training and testing.
- The supervisor approves the invoice-only scope.

10.3 Dependencies

Supervisor approval of scope; access to an ERP sandbox or Odoo instance (Phase 12); dataset availability (Phase 8); GPU or cloud access (Phase 8).

11

Risk Analysis



Figure 11.1: *Initial risk matrix. Identifiers refer to the table below.*

ID	Risk	Likelihood	Impact	Mitigation
R1	Not enough labelled data	High	High	Synthetic data, templates, invoice-only scope.
R2	Custom OCR under-performs	Medium	High	Baseline comparison, error analysis, replaceable OCR interface.
R3	Package incompatibility	Medium	Low	Python 3.13, pinned versions.
R4	ERP API unknown / no Odoo access	Medium	High	Mock ERP and adapter interface; ask for access early.
R5	Sensitive data exposure	Low	High	Synthetic data, private storage, minimal logging.
R6	Duplicate ERP records	Low	Medium	Idempotency keys and retry tests.
R7	Scope creep	High	Medium	Written scope; extras go to future work.
R8	Microservice overhead	Medium	Medium	Modular monolith first.
R9	OCR job crashes	Medium	Low	Persisted status, bounded retries.

ID	Risk	Likelihood	Impact	Mitigation
R10	Training hardware or time lacking	Medium	High	Small model, cloud GPU, smaller dataset.

12

Project Plan

12.1 Phases and stages

The project has 18 phases in four stages: A Foundation (1–5), B Intelligence (6–10), C Application (11–13), D Delivery (14–18). The rule is to complete one end-to-end workflow before adding architectural complexity.

12.2 Proposed timeline

Assumption

> The 30-week timeline is a *proposal*. The real start date, deadline and supervisor checkpoints are not known and must replace these weeks.

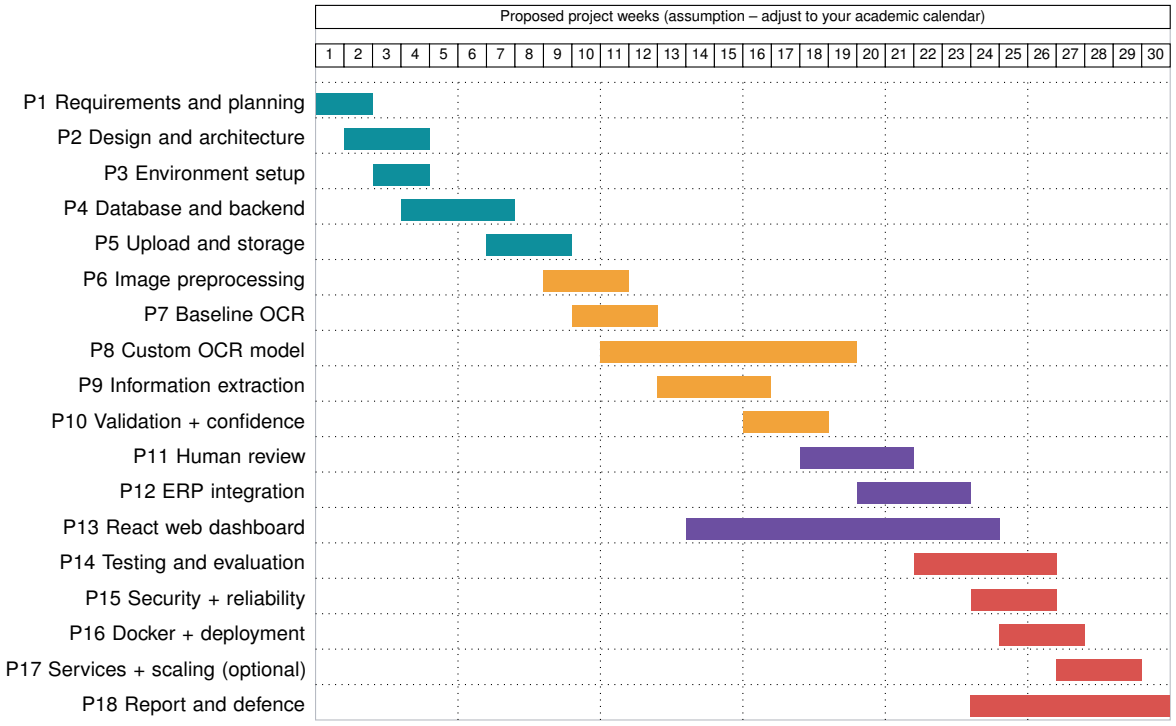


Figure 12.1: *Proposed Gantt chart (colours: teal = foundation, amber = intelligence, violet = application, red = delivery).*

12.3 Milestones

ID	Milestone	Proposed week	Evidence
M1	Requirements approved by supervisor	2	Signed Phase 1 document
M2	Backend + database + upload working	9	Passing tests, demo
M3	Baseline OCR results	12	CER/WER table
M4	Custom OCR trained and compared	19	Evaluation report
M5	End-to-end workflow with mock ERP	23	Recorded demo run
M6	Evaluation and security complete	26	Test reports
M7	Deployed system and final report	30	Report, slides, demo

12.4 Definition of done

A phase is done when: its deliverable exists, automated tests for it pass, documentation is updated, and the result is demonstrable to the supervisor.

13

Traceability Matrix

Each objective is linked to requirements and to the test that will prove it.

Obj.	Requirements	Verified by	Phase
O1	FR-03, 04, 05, 06	API and upload tests	5, 14
O2	FR-07	Preprocessing A/B comparison	6
O3	FR-08, NFR-07	CER/WER experiment	7, 8
O4	FR-09	Field-level F1 evaluation	9
O5	FR-11, FR-17	Review-routing evaluation	10
O6	FR-10	Seeded-error unit tests	10
O7	FR-12, 16, 19, 22, NFR-08	Usability check, UI tests	11, 13
O8	FR-13, NFR-04	Audit-log tests	4, 11
O9	FR-14, 15, 20, 21	Mock-ERP integration tests	12
O10	NFR-06, NFR-09, NFR-10, FR-18	Clean-machine setup, health checks	16, 17

14

Open Decisions and Sign-off

14.1 Decisions needed to close Phase 1

No.	Decision	Current proposal	Owner
D1	Final project name	DocuMind AI (working name)	Student
D2	Supervisor name and checkpoints	To be completed	Student / supervisor
D3	Prototype document type	Invoices only	Supervisor
D4	ERP target	Mock ERP first, Odoo later	Student / supervisor
D5	Odoo version and access	Unknown	Student
D6	Dataset source and size	Synthetic + authorised samples; size after baseline	Student / supervisor
D7	Training hardware	Local GPU or cloud notebook	Student
D8	Real calendar and deadlines	Replace proposed 30 weeks	Student
D9	Policy when a reviewer overrides a validation error	Allow with mandatory reason	Supervisor

14.2 Phase 1 completion checklist

- ☐ Problem, objectives and scope reviewed with the supervisor.
- ☐ Decisions D1–D9 answered.
- ☐ Functional and non-functional requirements accepted.
- ☐ Provisional targets agreed as planning values.
- ☐ Timeline replaced by the real academic calendar.
- ☐ Cover page completed.

14.3 Approval

Student	Tuheu Tchoubi Pempeme Moussa Fahdil	Date:
Supervisor	[Supervisor name – to be completed]	Date: